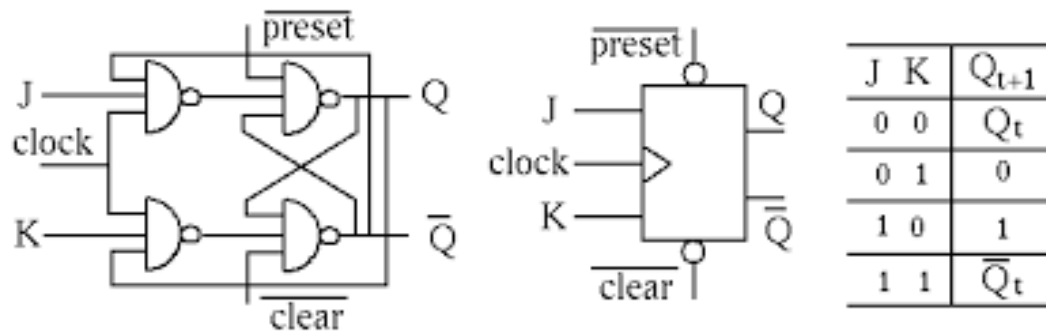
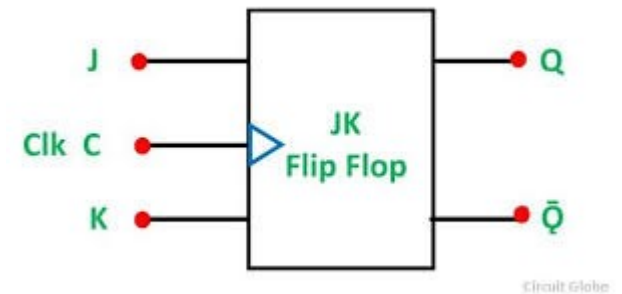


3 bit Ripple Counter using JK Flipflop

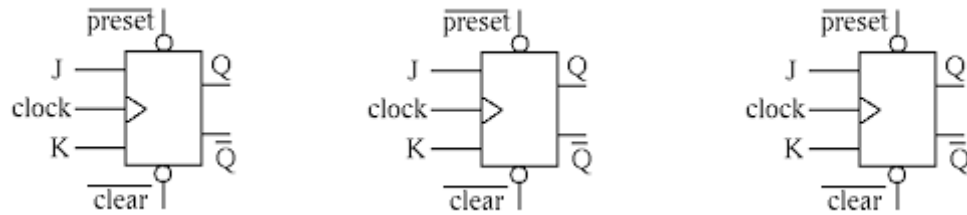
QA	QB	QC	QA'	QB'	QC'
0	0	0	1	1	1
0	0	1	1	1	0
0	1	0	1	0	1
0	1	1	1	0	0
1	0	0	0	1	1
1	0	1	0	1	0
1	1	0	0	0	1
1	1	1	0	0	0

$0 \text{ to } 2^N - 1 \rightarrow \text{Up Count}$

$2^N - 1 \text{ to } 0 \rightarrow \text{Down Count}$



3 Bit Asynchronous counter



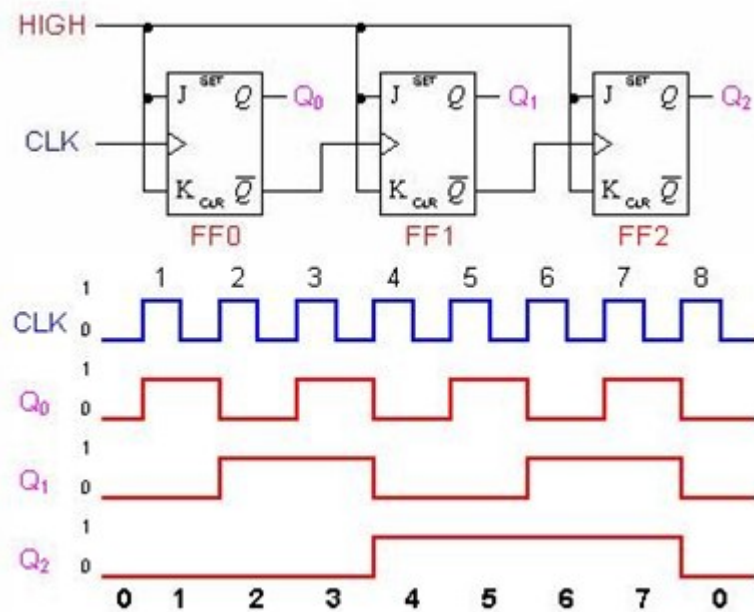
QA	QB	QC	QA'	QB'	QC'
0	0	0	1	1	1
0	0	1	1	1	0
0	1	0	1	0	1
0	1	1	1	0	0
1	0	0	0	1	1
1	0	1	0	1	0
1	1	0	0	0	1
1	1	1	0	0	0

Example: modify the above counter for following state diagram

$0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 0$

3 bit Asynchronous counter:

Identify the state diagram

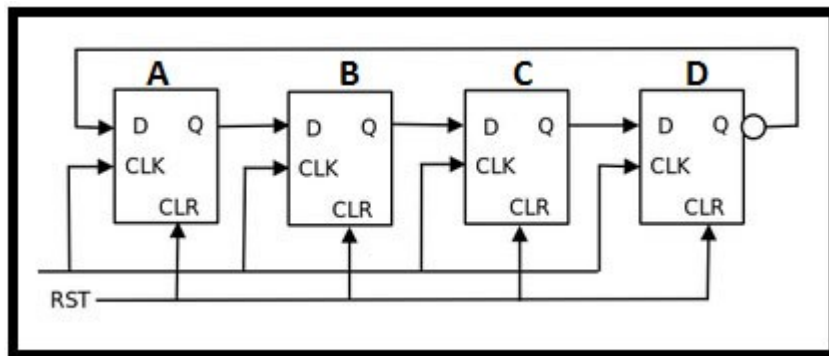


Ring and Johnson Counter

Two most important types of shift register counters are Johnson counter and Ring counter. These shift register counters with serial outputs are connected to serial inputs to produce particular pattern of sequences. These shift registers are used as counters because of the specified sequence of states.

Ring Counter

Ring counter is a basic application of shift registers. It is formed by the feedback of the output to its own input. This counter has N states where N denotes the number of [flip-flops](#) in the ring counter.



Q_A	Q_B	Q_C	Q_D
1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1

A 4 bit ring counter circuit is shown in the figure above. It consists of 4 D-flipflops, FFA, FFB, FFC and FFD. Each of these D flip-flops has an input D and output Q. At first, a **CLEAR** signal is applied to the flip-flops to RESET the outputs to zero. Then a **PRESET** pulse is applied to the flip-flop FFA before the clock pulse is given. This step allows putting the value '1' to the ring counter circuit. When each time the clock pulse is given, the counter circulates the data among all the four flip-flops. Modulo-4 or mod-4 counter is the popular name of 4 bit ring counter. Each output value of this counter has a frequency $\frac{1}{4}$ th of the main frequency value.

Advantages

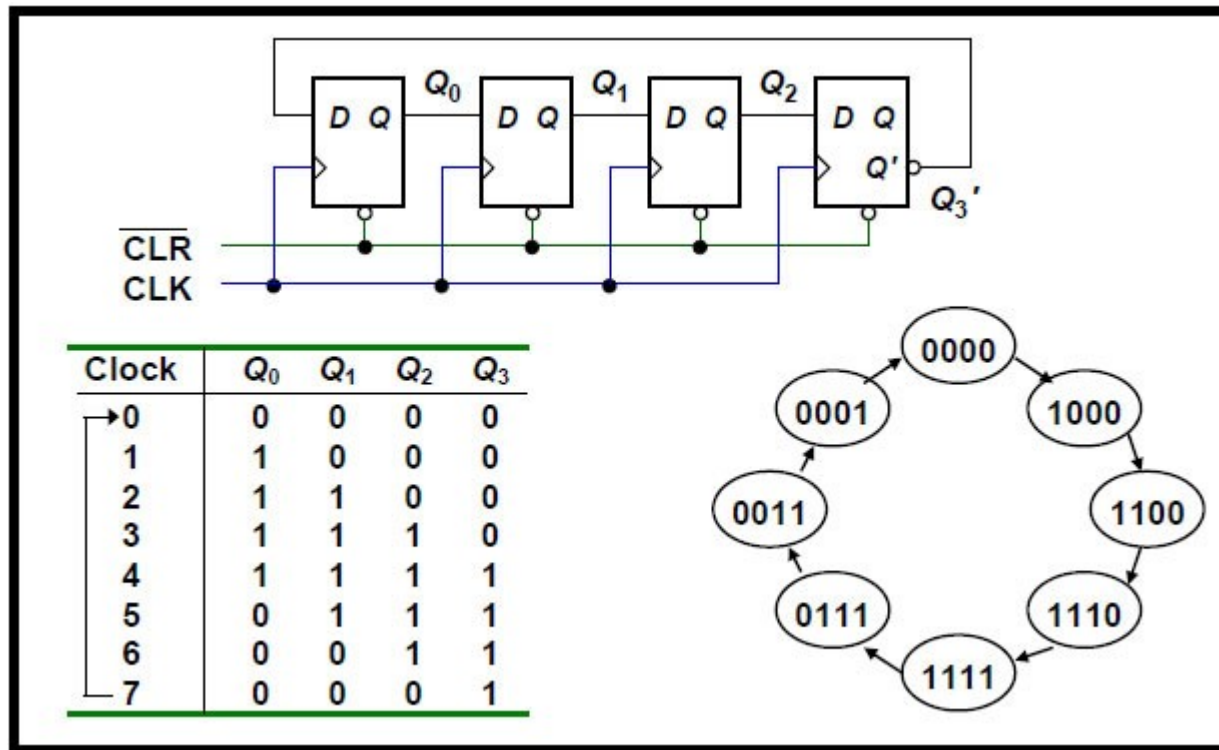
- Can be implemented using D and JK flip-flops.
- It is a self-decoding circuit.

Disadvantages

- Only four of the 15 states are being utilized.

Johnson Counter

Johnson Counter also known as **Twisted Ring Counter** is another basic application of shift registers with a feedback. Here the feedback is given from the inverted output of the last flip flop to the input of the first flip-flop. Figure below shows a 4-bit Johnson counter. It consists of four flip-flops FF0, FF1, FF2 and FF3. Here the inverted output of the last flip-flop FF3 is given as feedback to the input of the first flip-flop FF0. Here, at first four logic zeros will be passed to the flip-flops. When clock pulses are given "1000", "1100", "1110", "1111", "0111", "0011", "0001", "0000" outputs will be obtained and the sequence will repeat for the next clock pulses.



Advantages

- More outputs than ring counter.

Disadvantages

- Only 8 of the 15 states are being used.

Reference: <https://www.mepits.com/tutorial/291/basic-electronics/ring-and-johnson-counter>